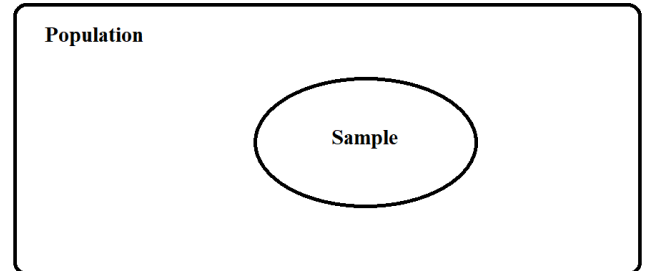


Collecting Data

In order to better understand the characteristics of a *population*, statisticians and researchers often use a *sample* from that population and make inferences based on the summary results from the sample.

Population - ENTIRE GROUP WE
WANT INFORMATION ABOUT



A population can be huge like “all women” or small like “top 200 grossing movies in 2011”.

Sample - A SUBSET OF THE POPULATION WE EXAMINE
TO GATHER INFORMATION ABOUT THE POPULATION.
The size of the sample can vary and depends on several factors we will examine throughout this course.

Census - COLLECTS DATA FROM EVERY INDIVIDUAL OF A POPULATION

The Census you have heard the most about is the one done every 10 years in the USA.

The way we collect data influences what we can and cannot say about a population.

Observational Study - OBSERVES INDIVIDUALS AND VARIABLES OF
INTEREST (DOESN'T ATTEMPT TO INFLUENCE RESPONSES)

- In an observational study, treatments are not imposed.
- Investigators examine data for a sample of individuals (retrospective) or follow a sample of individuals into the future collecting data (prospective) in order to investigate a topic of interest about the population.
- A sample survey is a type of observational study that collects data from a sample in an attempt to learn about the population from which the sample was taken.

Experiment - DELIBERATELY IMPOSES "TREATMENT" ON INDIVIDUALS
TO MEASURE THEIR RESPONSES.

We will discuss experiments in greater detail later in the unit. For now, let's discuss the various ways we plan an observational study.

It is only appropriate to make generalizations about a population based on samples that are randomly selected or otherwise representative of that population.

- A sample is only generalizable to the population from which the sample was selected.
 - Example, if you poll a sample of women asking about their shopping habits, you cannot take that result and apply it to men, since they were not represented in the sample that was taken.
- It is not possible to determine causal relationships between variables using data collected in an observational study.
 - While we observe variables and gather data in an observational study, we cannot make inferences between the variables. As long as it is a well-designed observational study, we can only apply the findings from the sample to the population.

Sampling

Because we make inferences about a population from the sample, it is very important that the sample is collected appropriately and that it is *representative* of the population being studied. The following is a list of possible sample designs and some of the advantages and disadvantages of each.

1) **Convenience sampling** – USES SUBJECTS THAT ARE READILY AVAILABLE.

Advantage: Easy and less costly to collect

Disadvantage: Not representative of the population

Example: In order to get an idea of how students think of the new school policy, the principal stands outside the library and asks a few students their opinions.

2) **Voluntary Response Sample** – SUBJECTS ARE ALLOWED TO DECIDE IF THEY WANT TO RESPOND. (A.K.A. Self –selected survey)

Advantage – Easy to collect

Disadvantage – Over represents people with strong opinions.

Example: After the State of the Union speech, ABC tells its audience to call a 1-900-555-1234 if they thought the speech was good and 1-900-555-7890 if they thought the speech was bad (there is a \$0.50 charge for the call).

3) **Simple Random Sample (SRS)** – CONSISTS OF "n" INDIVIDUALS FROM THE POPULATION CHOSEN IN A WAY SO THAT EVERY SET OF "n" INDIVIDUALS FROM THE POPULATION HAS AN EQUAL CHANCE OF BEING SELECTED.

This is often the best and most appropriate way to collect data for a sample.

Advantages – Easy to accomplish using a table of random digits; likely to produce samples that are good representatives of the population.

Disadvantage – None (cost or time could be an issue)

Example: In order to determine how happy students are with their education at BHS, the principal assigns each student a number from 1 to 1230 (the number of students at the school) and then uses a random number generator to choose 50 numbers between 1 and 1230. He then surveys all the students with the chosen numbers.

4) **Stratified random sampling** – DIVIDE POPULATION INTO SIMILAR GROUPS OF INDIVIDUALS (STRATA), SELECT A SRS FROM EACH STRATA AND COMBINE THEM FOR YOUR TOTAL SAMPLE.

Advantage: Can produce more exact information (especially in large populations) by taking advantage of the fact that individuals in the same strata are similar to one another.

Disadvantage: Not appropriate unless strata are easily defined.

Example: In order to get a better idea of what BHS athletes thought about homecoming last year, the director divides all BHS athletes into the teams they play for, and then selects a random sample from each sports team. His full sample consists of aggregating the random samples from each team.

5) **Cluster Sampling** – DIVIDE THE POPULATION INTO SECTIONS OR "CLUSTERS" THEN RANDOMLY CHOOSE A FEW CLUSTERS. EVERY INDIVIDUAL IN THOSE CLUSTERS BECOMES PART OF YOUR SAMPLE.

Advantage – Don't need a list of entire population

Disadvantage – More variability between samples depending on how clusters are determined.

Example – A psychologist at the University of Pennsylvania collects a sample by first dividing up the students into their respective schools (Wharton, engineering, nursing, arts and sciences) then by the departments that their major is in, and then she selects a few departments at random and surveys every student within those chosen departments.

6) **Systematic random sampling** – RANDOM STARTING POINT THEN SELECTING EVERY K^{th} MEMBER OF THE POPULATION

Advantage: Every member has an equal probability of being selected

Disadvantage: Not every sample of size n has an equal chance of being selected

Example: HP Selects every 200th computer off the assembly line and inspects it for quality control.

When an item from a population can be selected only once, this is called WITHOUT REPLACEMENT.

When an item from the population can be selected more than once, this is called WITH REPLACEMENT.

There are advantages and disadvantages for each sampling method depending upon the question that is to be answered and the population from which the sample will be drawn.